

Inequalities – 1

The Cauchy–Bunyakovsky inequality as a technique

Homework

10 сентября 2026 года

Theoretical minimum. The discrete, integral and inner product forms and the proof through the discriminant of $\int (f - \lambda g)^2$; the equality case as proportionality; the Engel form $\sum a_i^2/b_i \geq (\sum a_i)^2/\sum b_i$; the choice of the splitting $h = fg$ dictated by the equality case; the Gram determinant.

1. (RMO 1997) Prove that $\int_{-1}^1 f^2 \geq \frac{1}{2}(\int_{-1}^1 f)^2 + \frac{3}{2}(\int_{-1}^1 xf)^2$ for every continuous f , and identify the functions for which equality holds.
2. (Putnam 2004) Let $f(x, y)$ be continuous on the unit square. Show that $\int_0^1 (\int_0^1 f(x, y) dx)^2 dy + \int_0^1 (\int_0^1 f(x, y) dy)^2 dx \leq (\int_0^1 \int_0^1 f)^2 + \int_0^1 \int_0^1 f^2$.
3. (Putnam 2013) For continuous f on $[0, 1]$ put $\mu(f) = \int_0^1 f$, $\text{Var}(f) = \int_0^1 (f - \mu(f))^2$ and $M(f) = \max_{[0,1]} |f|$. Show that $\text{Var}(fg) \leq 2 \text{Var}(f)M(g)^2 + 2 \text{Var}(g)M(f)^2$.
4. (RMO 2008) Let $f \in C^1[0, 1]$ with $f(\frac{1}{2}) = 0$. Prove that $\int_0^1 (f')^2 \geq 12(\int_0^1 f)^2$.
5. (RMO 2025) Let f be differentiable on $[0, 1]$ with integrable derivative and $f(1) = 0$. Prove that $\int_0^1 (xf'(x))^2 dx \geq 12(\int_0^1 xf(x) dx)^2$, and compare the weight with the previous problem.
6. (IMC) Let (x_i) be a decreasing sequence of positive numbers.
 - a) Show that $(\sum_{i=1}^n x_i^2)^{1/2} \leq \sum_{i=1}^n \frac{x_i}{\sqrt{i}}$.
 - b) Show that there is a constant C with $\sum_{m=1}^{\infty} \frac{1}{\sqrt{m}} (\sum_{i=m}^{\infty} x_i^2)^{1/2} \leq C \sum_{i=1}^{\infty} x_i$.

7. (Петрополитана 2024) Prove that $\int_0^1 e^f \cdot \int_0^1 e^{-f} \geq 1 + \int_0^1 f^2 - \left(\int_0^1 f\right)^2$ for every continuous $f: [0, 1] \rightarrow \mathbb{R}$.
8. (ФКН, весна 2023) Let $f: \mathbb{R} \rightarrow \mathbb{R}_+$ be continuous with period T . Prove that $\int_0^T \frac{f(x)}{f(x+\alpha)} dx \geq T$ for every real α .
9. Prove that the Gram determinant of any finite family of vectors in an inner product space is non-negative, deduce the Cauchy–Bunyakovsky inequality from the case of two vectors, and interpret the general case as a statement about volumes.
10. (SEEMOUS 2018) Let $f: [0, 1] \rightarrow (0, 1)$ be Riemann integrable. Show that $2 \int_0^1 x f^2(x) dx \cdot \int_0^1 (f^2(x) + 1) dx < \int_0^1 f^2(x) dx \cdot \int_0^1 f(x) dx$.